

PC-01 · SURFACTANTS · PERSONAL CARE

HS2000

Decyl Glucoside

INCI	CAS	EC	FAMILY	SUPPLIED AS
Decyl Glucoside	54549-25-6	259-218-1	PC-01	Aqueous solution, ~50% active

01 POSITIONING

A renewable, mono-constituent surfactant with the dossier the next regulatory cycle will demand — while legacy ethoxylates accumulate restrictions.

02 HOW IT WORKS

01 Sugar head, fatty tail

The glucoside head brings water affinity without ethoxylation; the fatty chain drives adsorption at the interface and lowers surface tension fast.

02 Mildness by design

Non-ionic character means low interaction with skin proteins — the base of its mildness profile in cleansing systems.

03 Foam architecture

Chain length tunes the foam: shorter chains wet and flash-foam; longer chains build dense, creamy, stable lather.

03 APPLICATIONS

- Sulfate-free shampoos and shower gels
- Baby care and syndet bars
- Facial cleansers and micellar waters
- Hard-surface and I&I cleaners
- Co-surfactant to boost mildness of anionic bases

04 FORMULATION GUIDE

USE LEVEL	2–20%
PH & CONDITIONS	Stable pH 3–12; formulate rinse-off at 5.0–6.5

- Add to the water phase under moderate stirring; avoid prolonged high shear to control foam in the vessel.
- Combine with betaines or sarcosinates to build viscosity and creaminess in sulfate-free systems.

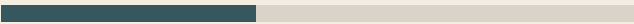
05 TYPICAL SPECIFICATION

Appearance	Clear to yellowish viscous liquid
Solid content	≥ 50%
pH (10% solution)	11.5–12.5
Residual alcohol	≤ 1.0%

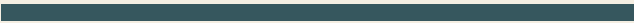
Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Surface tension at 0.1%
APG class · ~28–30 mN/m



Biodegradability (OECD)
APG class · readily biodegradable



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.